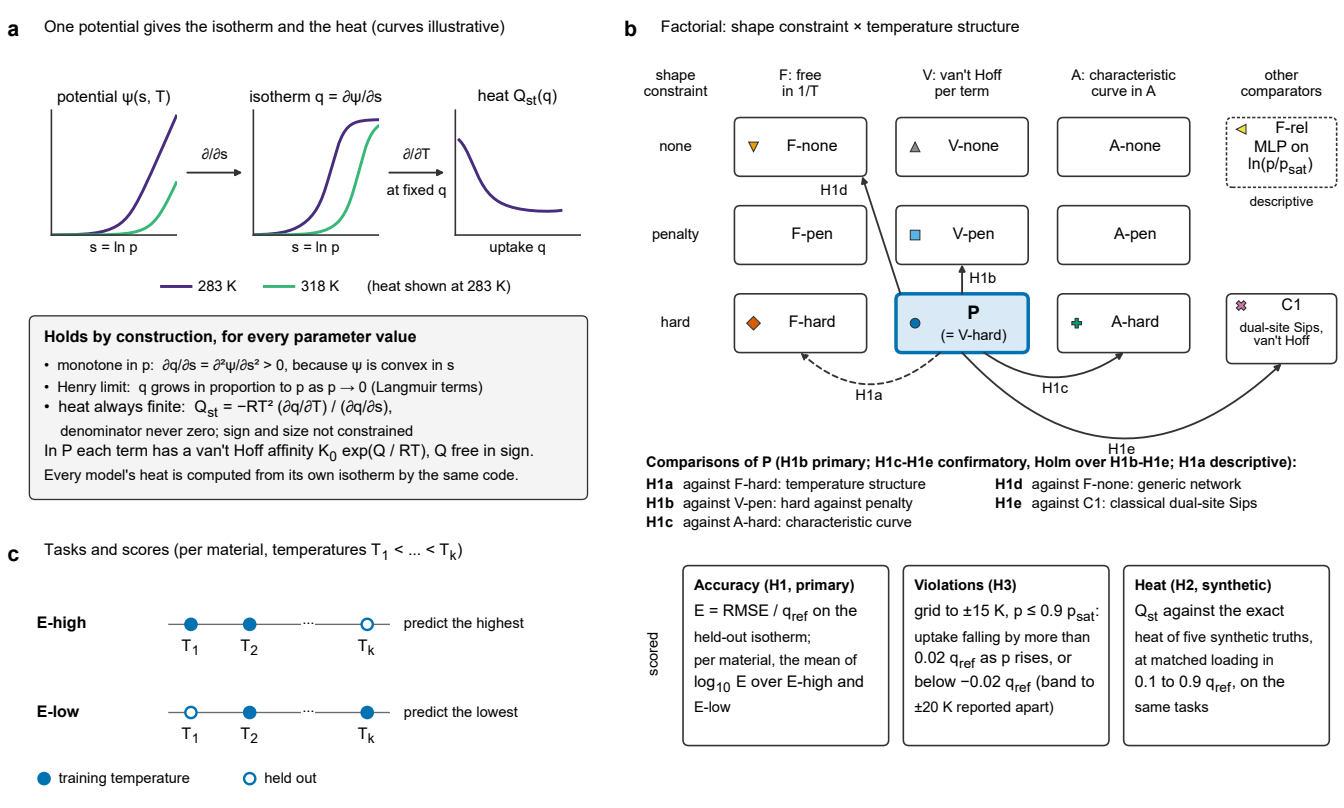

a
One potential gives the isotherm and the heat (curves illustrative)
potential ψ(s, T)
isotherm q = ∂ψ/∂s
heat Q_st(q)
∂/∂s
∂/∂T
at fixed q
s = ln p
s = ln p
uptake q
283 K
318 K
(heat shown at 283 K)
Holds by construction, for every parameter value
• monotone in p: ∂q/∂s = ∂²ψ/∂s² > 0, because ψ is convex in s
• Henry limit: q grows in proportion to p as p → 0 (Langmuir terms)
• heat always finite: Q_st = −RT² (∂q/∂T) / (∂q/∂s),
denominator never zero; sign and size not constrained
In P each term has a van't Hoff affinity K_0 exp(Q / RT), Q free in sign.
Every model's heat is computed from its own isotherm by the same code.
b
Factorial: shape constraint × temperature structure
shape constraint
F: free in 1/T
V: van't Hoff per term
A: characteristic curve in A
other comparators
none
penalty
hard
F-none
V-none
A-none
F-rel MLP on ln(p/p_sat)
descriptive
H1d
F-pen
V-pen
A-pen
H1b
F-hard
P
(= V-hard)
A-hard
C1
dual-site Sips, van't Hoff
H1a
H1c
H1e
Comparisons of P (H1b primary; H1c-H1e confirmatory, Holm over H1b-H1e; H1a descriptive):
H1a against F-hard: temperature structure
H1b against V-pen: hard against penalty
H1c against A-hard: characteristic curve
H1d against F-none: generic network
H1e against C1: classical dual-site Sips
c
Tasks and scores (per material, temperatures T_1 < ... < T_k)
E-high
T_1
T_2
T_k
predict the highest
E-low
T_1
T_2
T_k
predict the lowest
training temperature
held out
scored
Accuracy (H1, primary)
E = RMSE / q_ref on the held-out isotherm; per material, the mean of log_10 E over E-high and E-low
Violations (H3)
grid to ±15 K, p ≤ 0.9 p_sat: uptake falling by more than 0.02 q_ref as p rises, or below −0.02 q_ref (band to ±20 K reported apart)
Heat (H2, synthetic)
Q_st against the exact heat of five synthetic truths, at matched loading in 0.1 to 0.9 q_ref, on the same tasks